

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution Affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
Due Date	27-01-2026		

Experiment 3: Regression Analysis using Linear and Regularized Models

Aim and Objective

To implement linear regression and regularized regression models for predicting a continuous target variable, evaluate their performance using multiple metrics, visualize model behavior, and analyze overfitting, underfitting, and bias–variance characteristics.

Dataset Description

A real-world regression dataset containing numerical and categorical features related to loan applications is used. Dataset contains records of the amount of loan that has been sanctioned to a user along with the features to predict the sanctioned amount. The target variable is the **loan amount sanctioned in US Dollars**.

Dataset reference:

- Kaggle: Predict Loan Amount Data

Preprocessing Steps

Numerical Features

1. **Simple Imputer:** To replace NaN values with mean of the corresponding column.
2. **MinMax Scaler:** To scale features to a uniform scale. Minmax scaler was chosen since many numerical features were skewed.

Categorical Features

1. **Simple Imputer:** To replace NaN values with mode of the corresponding column.
2. **Ordinal Encoder:** To encode categorical values to numerical values to be accepted by ML models in further steps.

Implementation Details

1. Load the dataset
2. Perform data preprocessing:
 - Handle missing values
 - Encode categorical variables
 - Standardize numerical features
3. Perform Exploratory Data Analysis (EDA)
4. Visualize feature distributions and target distribution
5. Split the dataset into training and testing sets
6. Train baseline Linear Regression
7. Train Ridge, Lasso, and Elastic Net models
8. Perform hyperparameter tuning using 5-Fold Cross-Validation
9. Evaluate all models using regression metrics

Visualisations

0.1 Target Variable Distribution

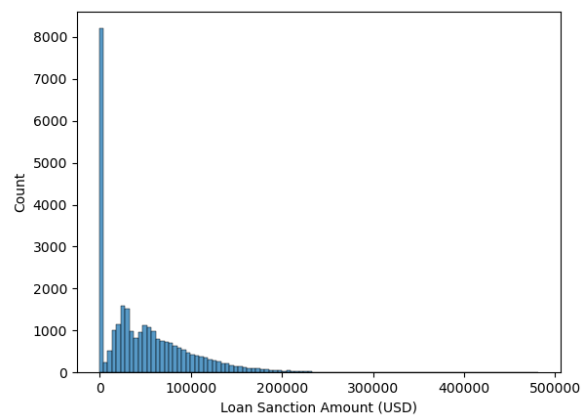


Figure 1: Histogram: Loan Amount USD

0.2 Target Variable Distribution

Task Description

Students must:

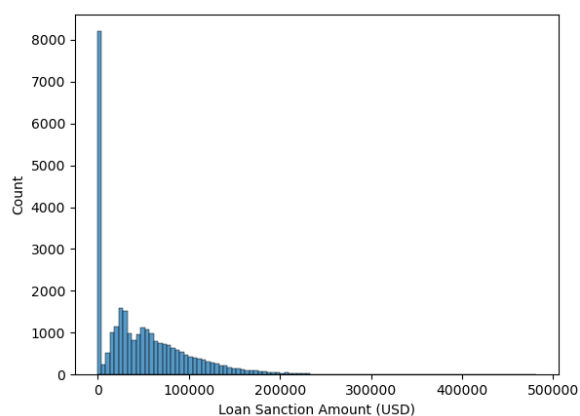


Figure 2: Histogram: Loan Amount USD

- Implement Linear, Ridge, Lasso, and Elastic Net regression models
- Tune regularization hyperparameters using Grid Search or Randomized Search
- Visualize regression results and errors
- Analyze overfitting, underfitting, and bias–variance trade-off

Hyperparameter Search Space

- Ridge: $\alpha \in \{0.01, 0.1, 1, 10, 100\}$
- Lasso: $\alpha \in \{0.001, 0.01, 0.1, 1, 10\}$
- Elastic Net:
 - $\alpha \in \{0.01, 0.1, 1, 10\}$
 - $l1_ratio \in \{0.2, 0.5, 0.8\}$

Hyperparameter Tuning Results

Table 1: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	Grid	$\alpha = 0.01$	0.5881
Lasso Regression	Grid	$\alpha = 10$	0.5912
Elastic Net Regression	Grid	$\alpha = 0.01$ $l1_ratio=0.8$	0.5900

Table 2: Cross-Validation Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	21505.482958	9.685427e+08	31114.931302	0.585436
Ridge Regression	21498.117466	9.620821e+08	31009.847353	0.588106
Lasso Regression	21578.193457	9.545941e+08	30886.197104	0.591210
Elastic Net Regression	21587.386713	9.574273e+08	30932.852548	0.590011

Cross-Validation Performance (K = 5)

Test Set Performance Comparison

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	24336.65	2534562674.15	50344.44	-0.116658
Ridge Regression	21926.093710	1017701e+09	31901.428436	0.551629
Lasso Regression	22117.575936	1.015707e+09	31870.163657	0.552507
Elastic Net Regression	21649.776432	1.015952e+09	21649.776432	0.552399

Effect of Regularization on Coefficients

Table 4: Coefficient Comparison

Feature	Linear	Ridge	Lasso	Elastic Net
Feature 1	-1.539e+03	-1.526641	-1.477e+03	-1.532e+03
Feature 2	5.378e+06	5.228e+04	0.000000	1.507e+03
Feature 3	3.556e+05	3.555e+05	3.386932	2.980e+05

Overfitting and Underfitting Analysis

Difference between training and validation errors

Effect of regularization strength

Improvement in generalization after tuning

Bias–Variance Analysis

- Bias behavior of Linear Regression
- Variance reduction using Ridge and Elastic Net

- Feature sparsity effect in Lasso

Conclusion

Summarize the performance of all regression models, justify the choice of optimal hyperparameters, and discuss the trade-off between accuracy and model complexity.

References

- Scikit-learn: Linear Models
- Scikit-learn: Hyperparameter Optimization
- Loan Amount Dataset